

Strike while the Iron is Hot:

Optimal Monetary Policy under State-Dependent Pricing

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- ▶ Optimal monetary policy is mainly studied in models, in which the frequency is held constant (Galí, 2008; Woodford, 2003)
- ▶ What does optimal monetary policy look like with endogenous variation in frequency?
How should CBs respond to a large inflation surge?

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 - ▶ Solve the non-linear model under perfect foresight using a new numerical algorithm
 - ▶ Contrast it to conventional results in Calvo (1983) time-dependent model

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 - ▶ Why? The Phillips curve is **non-linear**: it gets steeper as frequency increases.
 - ▶ Thus, stabilizing inflation is less costly in terms of output when inflation is high
- ▶ **Divine coincidence** holds for TFP shocks, either small or large.

Motivation: optimal policy in textbook Calvo framework

- New Keynesian Phillips Curve:

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa y_t, \quad \text{where } \kappa = \frac{(1 - \theta)(1 - \beta\theta)}{\theta},$$

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- ▶ Higher frequency raises the slope of the Phillips curve

$$\frac{\partial \kappa}{\partial (1 - \theta)} > 0$$

- ▶ Why? Prices are more flexible

Motivation: Optimal policy in textbook Calvo framework, cont.

- Second-order approximation of utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{\varepsilon}{\kappa} \pi_t^2 + y_t^2 \right]$$

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- ▶ How does optimal policy respond to temporary frequency increases in menu cost models?

Literature

- ▶ Nonlinear Phillips curve (Benigno and Eggertsson, 2023; Cerrato and Gitti, 2023)
 - ▶ Microfounded by state-dependent price setting
(Golosov and Lucas, 2007; Gertler and Leahy, 2008; Auclert et al., 2022)
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- ▶ Optimal policy in a menu cost economy
 - ▶ Optimal inflation target (Burstein and Hellwig, 2008; Adam and Weber, 2019; Blanco, 2021)
 - ▶ Small shocks, large shocks, optimal nonlinear target rule (comp. Galí, 2008; Woodford, 2003)
 - ▶ Focus on aggregate shocks (unlike Caratelli and Halperin, 2023, who study *sectoral* shocks)

Model

Overview of our state-dependent model

= Textbook New-Keynesian model with Calvo pricing (e.g. Galí, 2008)

- Calvo pricing

- + fixed costs of price adjustments η

→ state-dependent price setting

- + idiosyncratic, fat-tailed quality shocks $A_t(i)$

→ realistic price change distribution

- + intermediate goods in production (Basu, 1995)

→ realistic Phillips curve slope

= Heterogeneous-firm NK DSGE model.

Overview of the model

- ▶ Households: consume a Dixit-Stiglitz basket of goods, save and work

Overview of the model

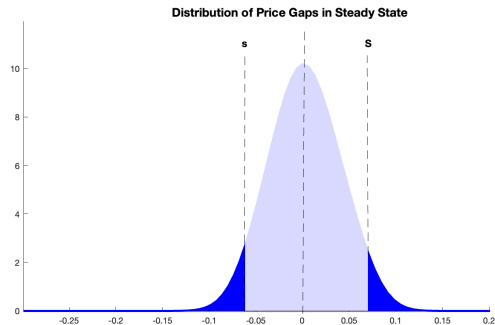
- ▶ Households: consume a Dixit-Stiglitz basket of goods, save and work
- ▶ Firms: produce differentiated goods using labor and the **final good** as inputs and are subject to aggregate TFP and CP shocks and **idiosyncratic “quality” shocks**. They have market power and set prices optimally subject to a **fixed cost** (Goloso and Lucas, 2007)

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- ▶ Monetary policy: maximizes household welfare under **commitment**

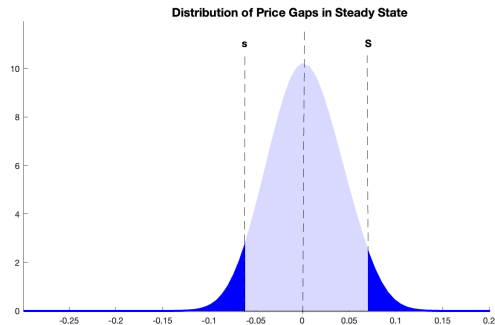
Model: Intuitive summary

- ▶ Each period, firm i chooses whether to reset its price and, if so, what new price to set
- ▶ The firm's optimality conditions define the reset price and the inaction region (S, s)



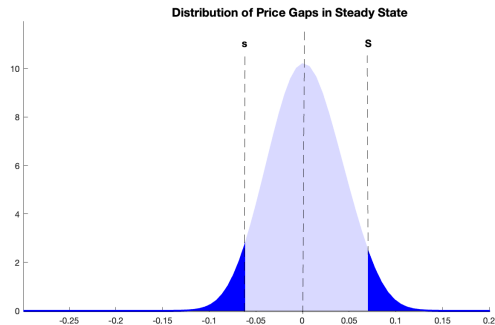
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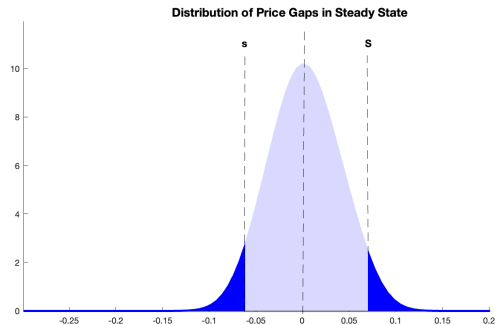
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- ▶ Let $p_t(i) \equiv \log(P_t(i)/(A_t(i)P_t))$ be the quality-adjusted log relative price



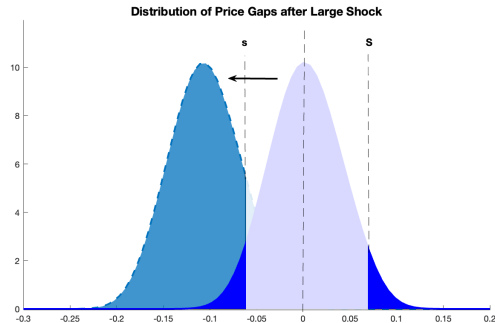
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- ▶ Let $p_t(i) \equiv \log(P_t(i)/(A_t(i)P_t))$ be the quality-adjusted log relative price
- ▶ Let $x_t(i) \equiv p_t(i) - p_t^*(i)$ be the difference of that price from the optimal price



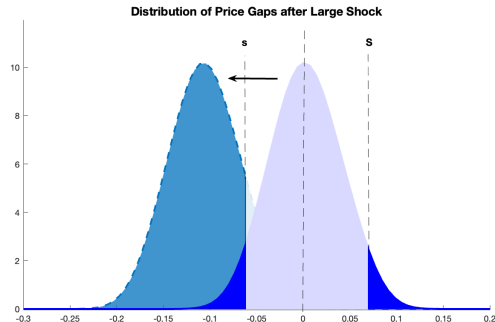
Model under large shock

- ▶ Large aggregate shock: shifts the distribution of price gaps for all firms
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Model under large shock

- ▶ Large aggregate shock: shifts the distribution of price gaps for all firms
- ▶ Limited impact on the (s, S) bands
- ▶ Pushes a large fraction of firms outside of the inaction region
- ▶ Large increase in the frequency of price changes and hence additional flexibility of the aggregate price level (on top of “selection”)



Optimal policy: simplified model

Inspecting the mechanism: A simplified model

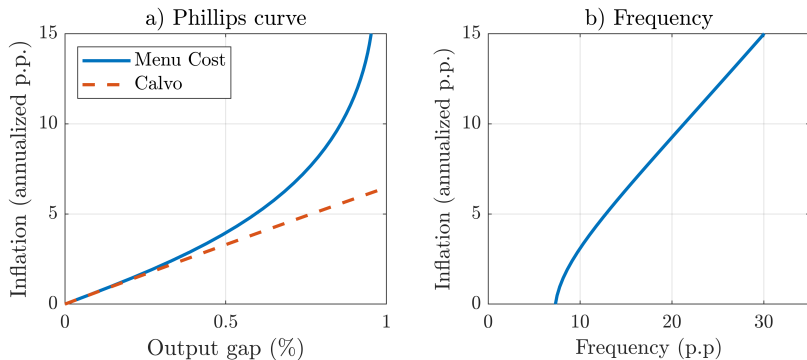
- ▶ In the Calvo model, we explain optimal policy as a choice of a planner selecting inflation and output to maximize quadratic welfare subject to the linearized Phillips curve
- ▶ To deliver a similar intuition but without linearization, we simplify the model by
 - ▶ assuming that prices are reset each night
 - ▶ muting roundabout and leptocurtic shocks
- ▶ Model becomes sequence of tractable static problems
- ▶ Simple expressions as function of π and y for
 - ▶ Phillips curve
 - ▶ Welfare

Planner's constraint: Phillips curve

- ▶ The model can be summarized by a single static equation relating inflation and output

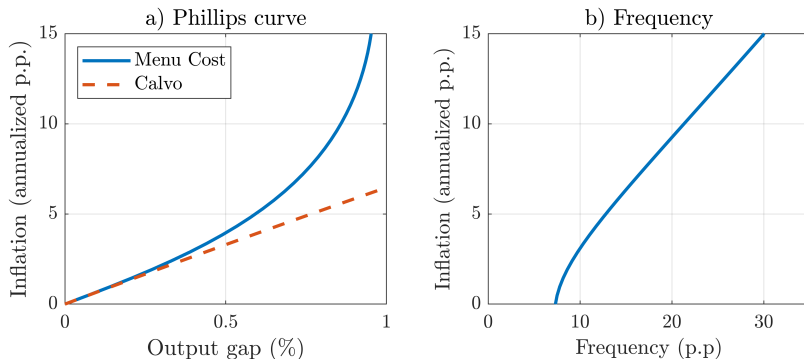
$$1 = \left[\int_s^S e^{(p)(1-\epsilon)} \phi \left(\frac{p + \pi}{\sigma} \right) dp + \left(\frac{\varepsilon(1-\tau)}{\varepsilon-1} Y \right)^{1-\epsilon} \left[1 - \int_s^S g^c(p + \pi) dp \right] \right],$$

Planner's constraint: Phillips curve



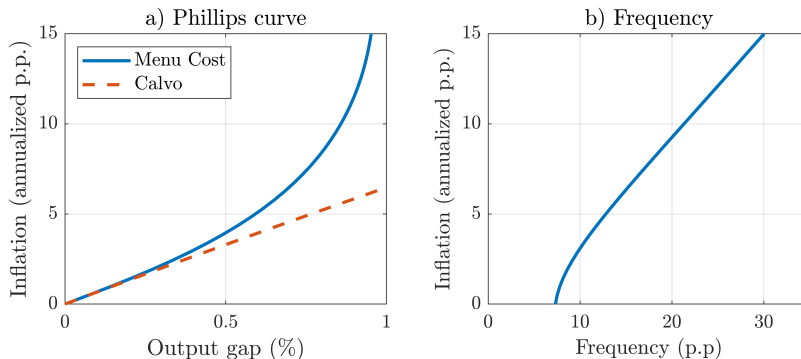
- Numerical calibration: match empirical frequency at normal times and at 10% inflation

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- ▶ Numerical calibration: match empirical frequency at normal times and at 10% inflation
- ▶ Phillips curve is nonlinear (close to linear under Calvo)

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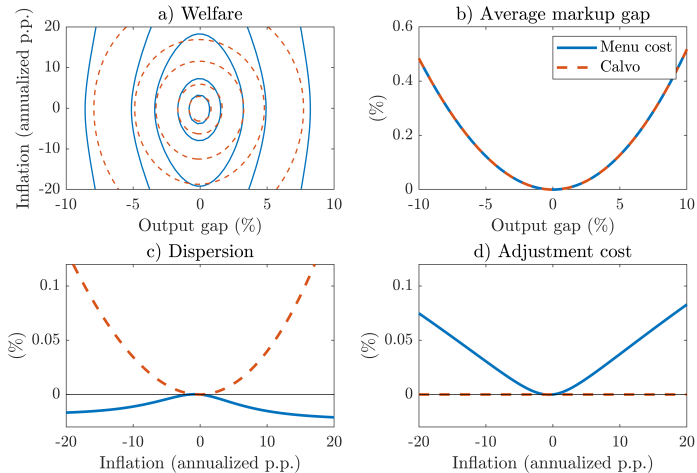
- ▶ Numerical calibration: match empirical frequency at normal times and at 10% inflation
- ▶ Phillips curve is nonlinear (close to linear under Calvo)
- ▶ Cheaper to disinflate at high inflation as price adjustments are more frequent

The planners objective: Welfare

- ▶ Welfare can be decomposed into three underlying distortions
- ▶ Each distortion depends on either output or inflation

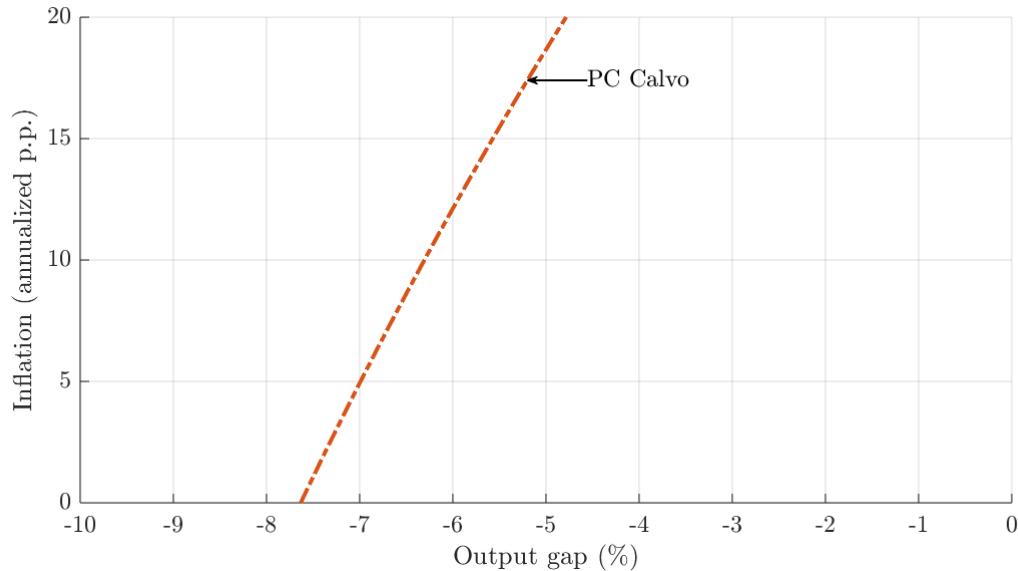
$$\begin{aligned} U - U^e = & \underbrace{\log(Y) - (Y - 1)}_{\text{Aggregate markup}} - \\ & \underbrace{Y \left(\int_s^S e^{p(-\epsilon)} \phi\left(\frac{p + \pi}{\sigma}\right) dp + \left(1 - \int_s^S \phi\left(\frac{p + \pi}{\sigma}\right) dp\right) e^{p^*(\pi)(-\epsilon)} - 1 \right)}_{\text{Markup dispersion}} - \\ & \underbrace{\eta \left[1 - \int_s^S \phi\left(\frac{p + \pi}{\sigma}\right) dp \right]}_{\text{Adjustment costs}} \end{aligned}$$

Planner's objective: Welfare

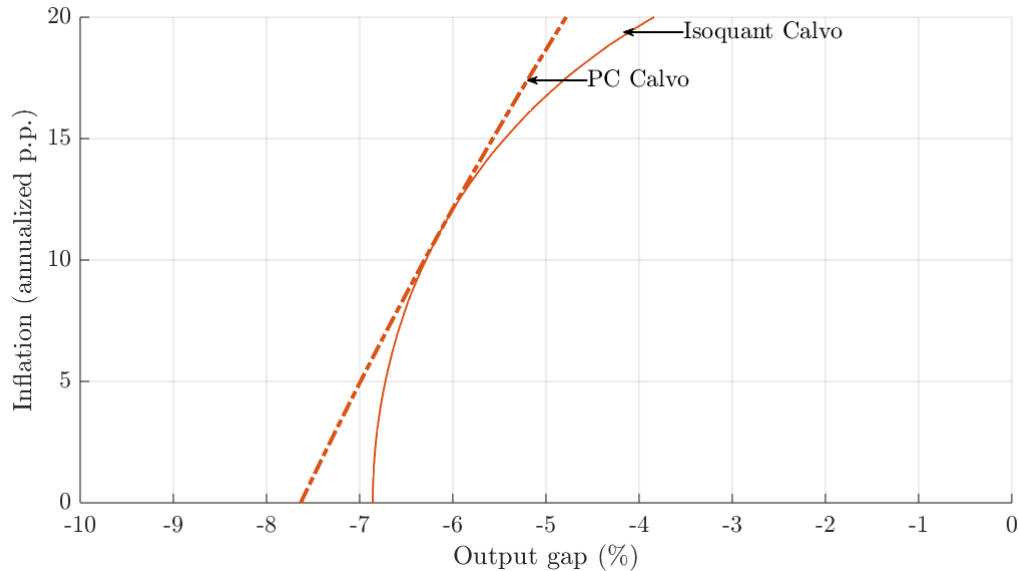


- Inflation is less costly in menu cost model, and for different reason

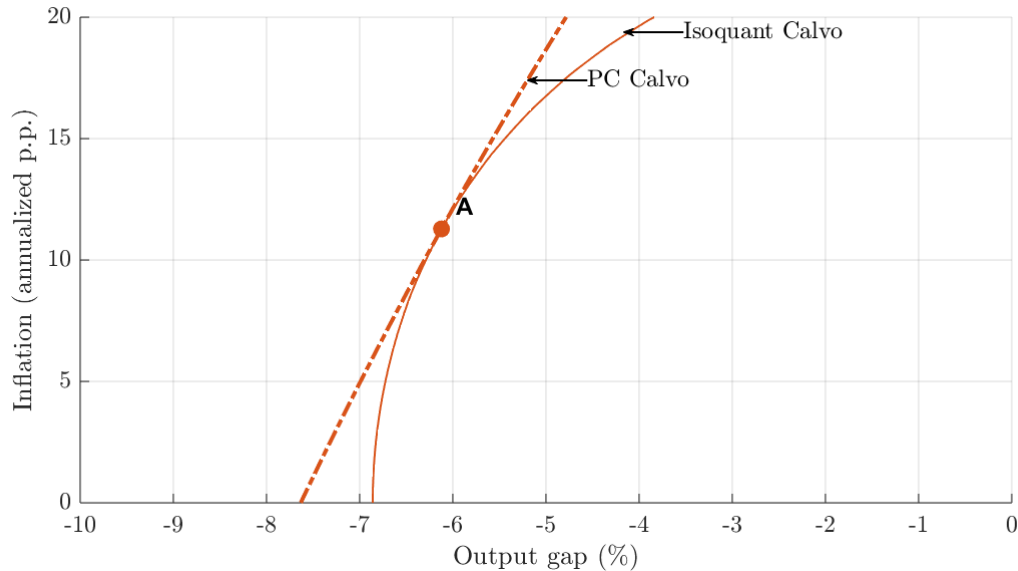
Optimal Policy



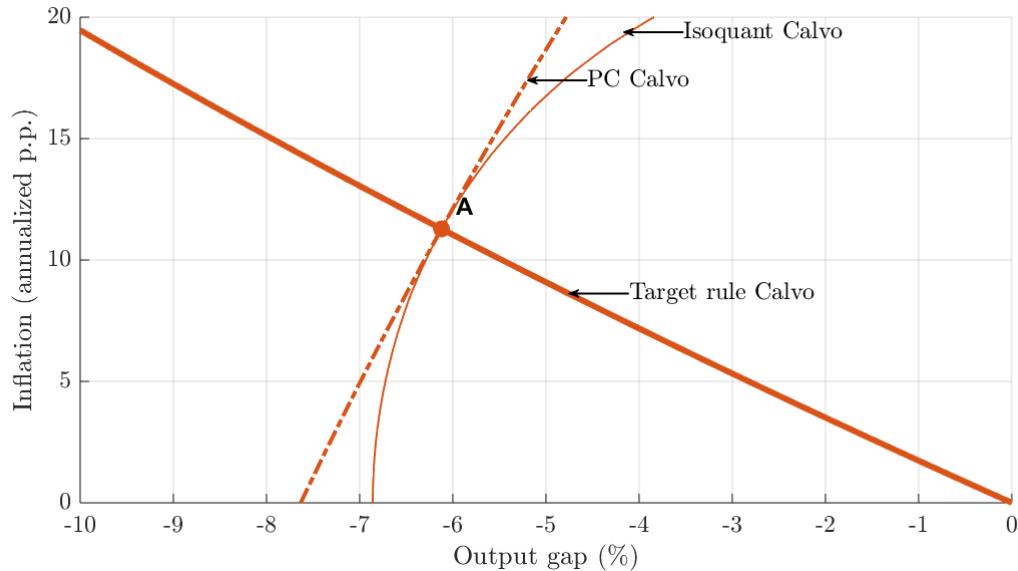
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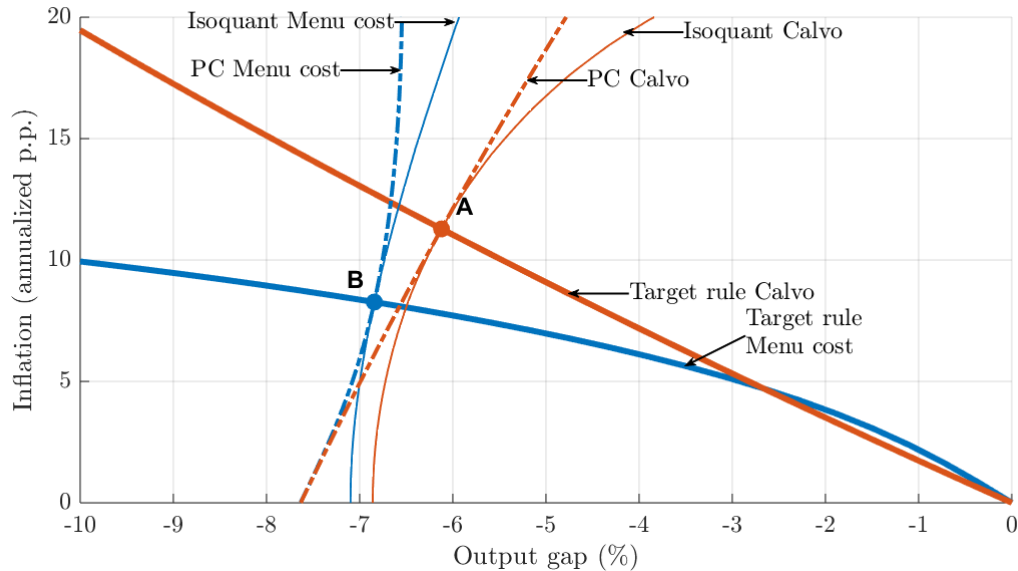
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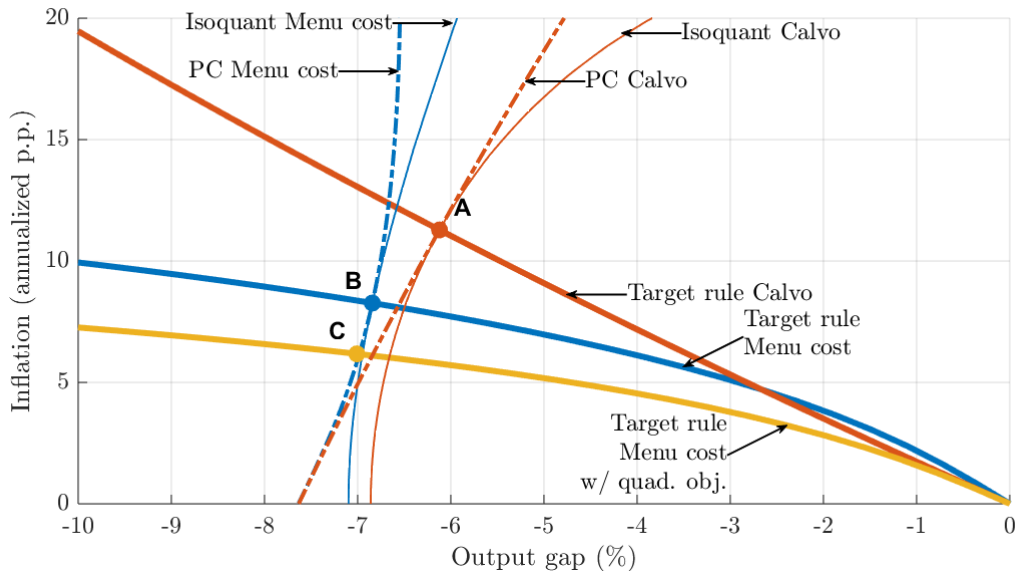
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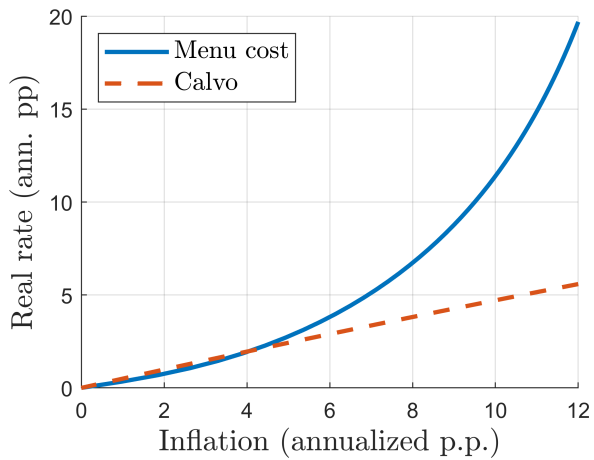
Optimal Policy



Optimal Policy



Optimal Policy Implementation



Optimal Policy

- ▶ Small shocks: Optimal policy similar to Calvo
- ▶ Large shocks: Optimal policy **strikes while hot** - Exploiting the lower sacrifice ratio, the planner stabilizes inflation more relative to output
- ▶ Reason: Convex Phillips curve and quasi-quadratic utility
- ▶ This is not obvious: A change in the Calvo parameter doesn't affect optimal policy in the Calvo model

Optimal policy: Full model

Full model: Calibration

	Baseline	Description	Reference
<i>Households and final producer</i>			
β	$0.99^{1/3}$	Discount rate	Golosov and Lucas (2007)
χ	1	Utility weight on labor	Set so that $w = C$
ϵ	6	Elasticity of substitution	Gagliardone et al. (2025)
<i>Differentiated goods producers</i>			
α	0.2	Labor share	Nakamura and Steinsson (2010)
η	0.064	Menu cost	Calibrated to match
σ	0.0306	Std dev. of quality shocks	moments of steady state
σ_1/σ_2	0.065	Ratio of low vs. high stdev	price change distribution
ϖ	0.934	Share of low volat. shock	at 2% inflation
<i>Aggregate shocks</i>			
ρ_A	$0.95^{1/3}$	TFP shock persistence	Smets and Wouters (2007)
ρ_τ	$0.9^{1/3}$	Cost-push persistence	Smets and Wouters (2007)

Full model: Calibration, cont.

	Steady state at 2% inflation			Phillips curve	10% inflation	
	Frequency	Size	Kurtosis	slope	Freq. incr.	Shock size
Data	8.7%	8.5%	4	0.006-0.106	14pp	
Baseline	8.7%	8.5%	4	0.08	14pp	-2.89%

► Fat-tailed idiosyncratic shocks

- The model is able to match the steady state frequency, size and kurtosis of price changes
- The model is able to capture large increase in frequency at 10% inflation (comp. Blanco et al. 2024)

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 - ▶ The model is able to match the steady state frequency, size and kurtosis of price changes
 - ▶ The model is able to capture large increase in frequency at 10% inflation (comp. Blanco et al. 2024)
- ▶ Roundabout production
 - ▶ The Phillips curve slope is high, but same order of magnitude as estimates
 - ▶ Shock size does not need to be extreme

Full model: Computation

- ▶ Challenges

- ▶ Price distribution $g_t(p_t)$ and value function $V_t(p_t)$ are **infinite-dimensional** objects
- ▶ We need sufficient accuracy for optimal policy assessment

Full model: Computation

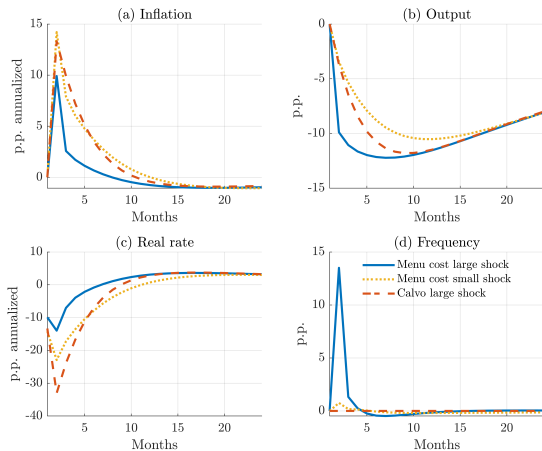
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- ▶ New algorithm, in discrete time

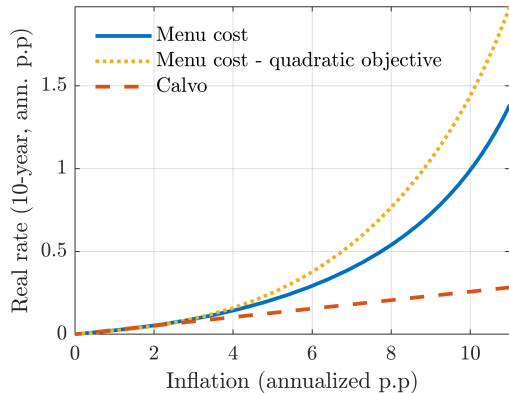
- ▶ Approximate distribution and value functions by piece-wise linear functions on grid.
- ▶ **Endogenous grid points**: (S,s) bands and the optimal reset price.
- ▶ Evaluate integrals analytically.
- ▶ Solve non-linearly in the **sequence space** using Dynare's perfect foresight Ramsey solver.

Full model: Strike while the iron is hot



- Policy stance gets over-proportionally hawkish as inflation increases

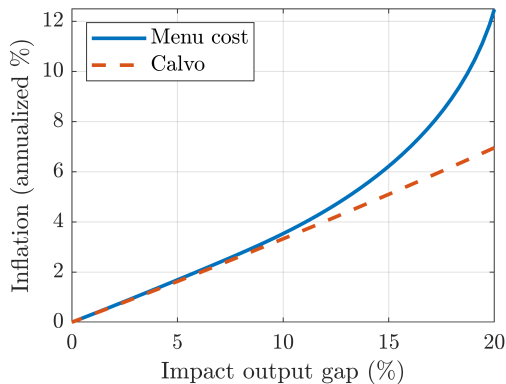
Full model: Strike while the iron is hot, cont.



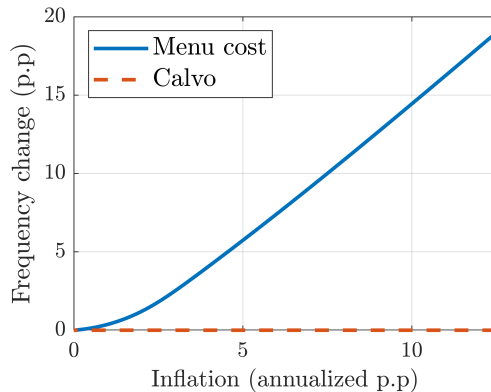
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Full model

Phillips relationship



Frequency



- Full model displays similar Phillips relationship and frequency as the simple model

Robustness

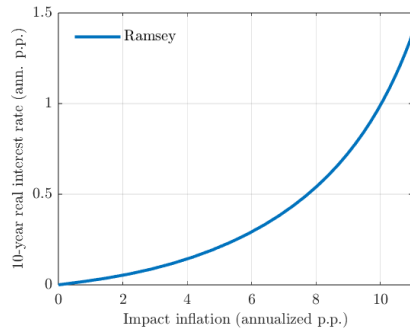
- ▶ The strike-while-the-iron-is-hot prescription is *qualitatively* robust to
 - ▶ alternative state-dependent pricing models
 - ▶ Calvo Plus (Nakamura and Steinsson, 2010)
 - ▶ Random Menu Cost (Dotsey et al., 1999; Gagliardone et al., 2025)
 - ▶ normal shocks and removing roundabout production
- ▶ Alternative models differ in the elasticity of frequency to inflation. If they are calibrated to this elasticity, the policy-stance inflation relationship is also *quantitatively* similar

Efficient shocks: “Divine coincidence” holds

- ▶ In the standard NK model with Calvo pricing: **divine coincidence** holds after shocks to TFP (A_t).
- ▶ Optimal policy stabilizes inflation and closes the output gap.
- ▶ Same result holds in menu-cost models, regardless shocks are small or large.

Policy implications

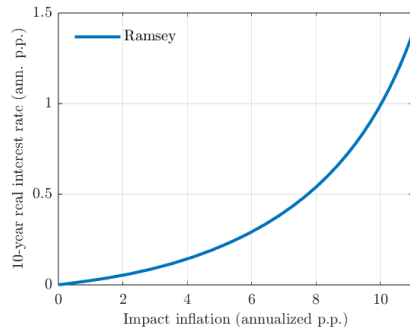
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Policy implications

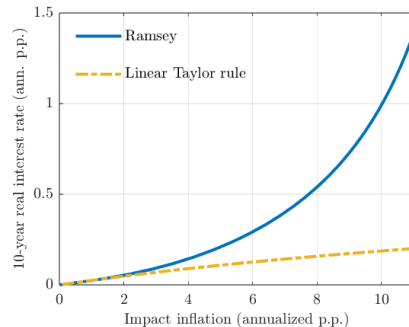
- ▶ How would a Taylor rule need to be modified to approximate optimal policy?
- ▶ Inflation coefficient needs to increase with inflation

$$\log R_t = \rho \log R_{t-1} + (1-\rho) \left\{ \left[\phi_\pi + \phi_F \pi_t^2 \right] \pi_t + \phi_y (y_t - y_t^*) \right\}$$



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- ▶ Pick ϕ_π, ϕ_F to match the policy-stance–inflation relationship (or, similarly, maximize welfare).

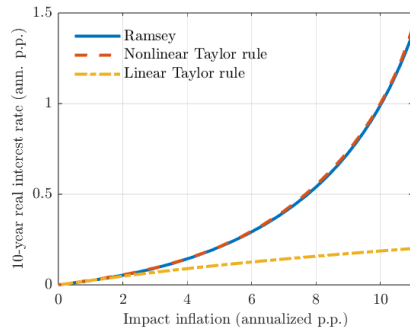
ρ	ϕ_π	ϕ_F	ϕ_y
0.9 ^{1/3}	2.22		0.5 ^{1/3}

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ρ	ϕ_π	ϕ_F	ϕ_y
$0.9^{1/3}$	2.22	135000	$0.5^{1/3}$

Policy implications, cont.

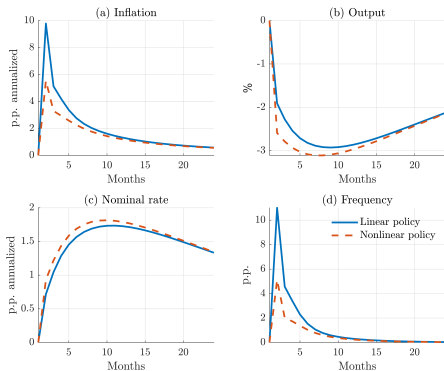
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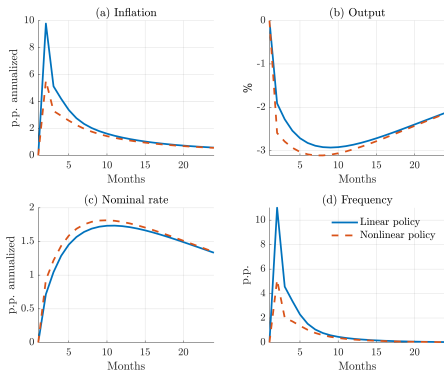
Policy implications, cont.

- ▶ Consider a policy maker that follows a linear rule ($\phi_F = 0$)
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- ▶ How would the outcome change if it followed the strike-it policy?



Policy implications, cont.

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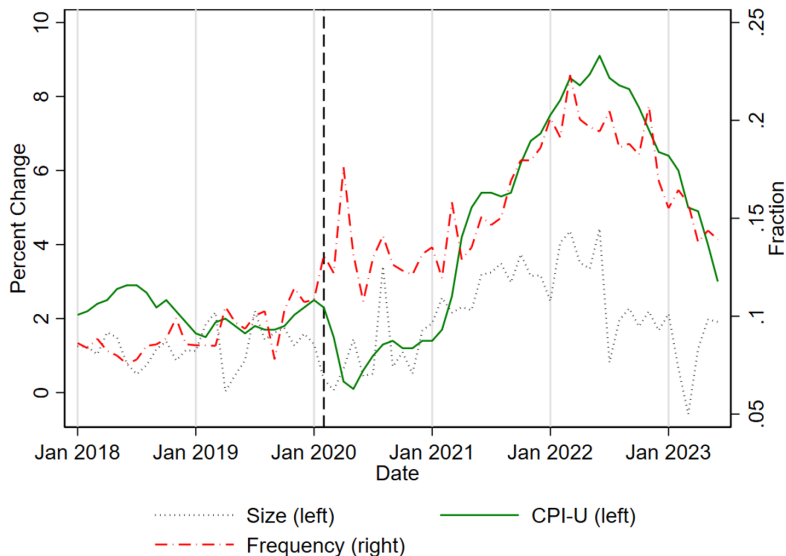
- ▶ Welfare gap relative to Ramsey 50% lower

Conclusion

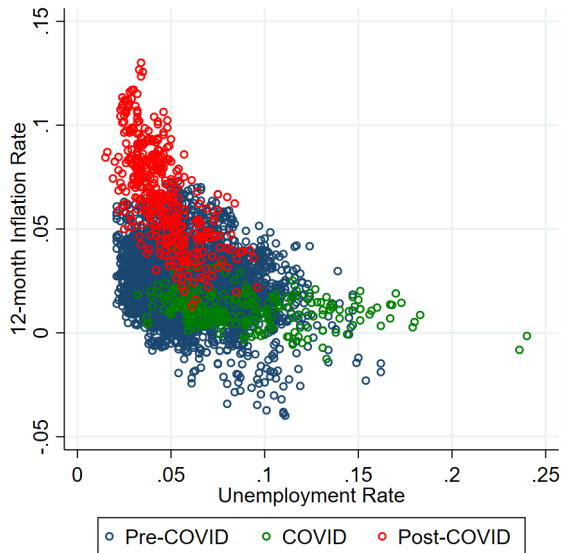
We study optimal policy in a menu cost model.

- ▶ The divine coincidence holds for TFP shocks.
- ▶ Optimal response to small cost-push shocks similar to [Calvo \(1983\)](#).
- ▶ More forceful response to large cost-push shocks: [strike while the iron is hot!](#)
- ▶ A taylor rule that responds more forcefully to inflation when inflation is high can approximate Ramsey policy.

CPI and frequency of price changes in the US, Montag and Villar (2023)



Phillips correlation across US cities, Cerrato and Gitti (2023)



Modified Phillips correlation time, Benigno and Eggertsson (2023)

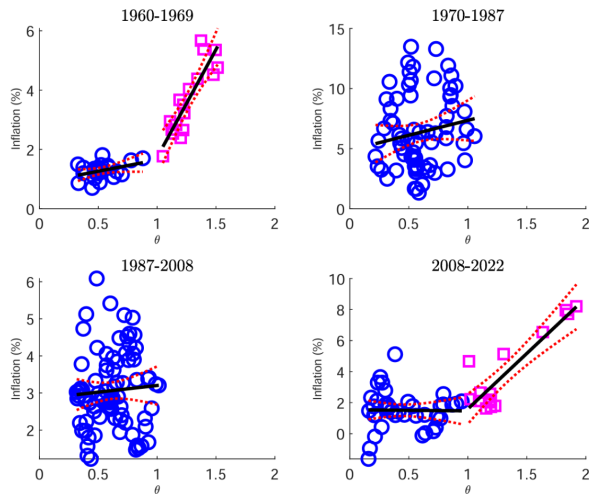
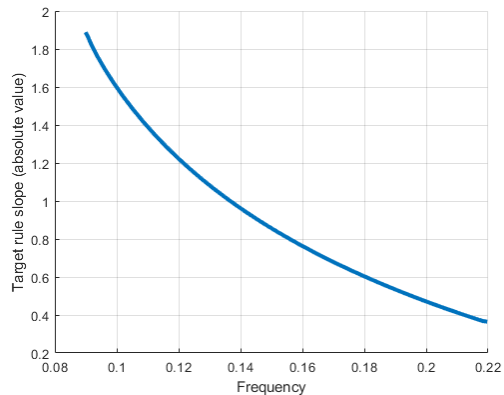


Figure 4: Inflation: CPI inflation rate at annual rates. θ : vacancy-to-unemployed ratio.

State-dependent inflation-output tradeoff

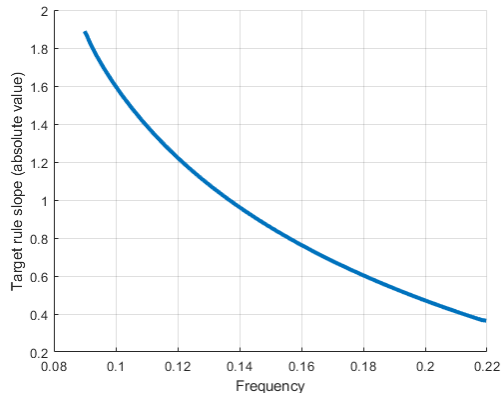
- Inflation-output tradeoff varies with frequency



State-dependent inflation-output tradeoff

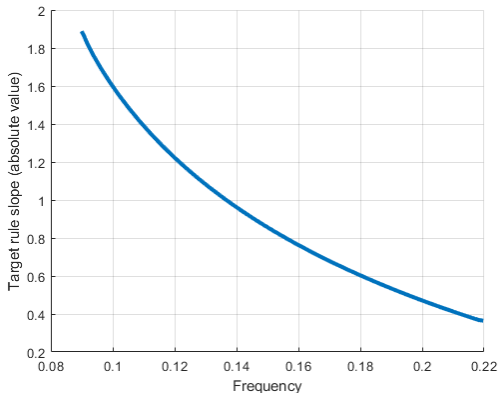
- ▶ Inflation-output tradeoff varies with frequency
- ▶ After large shocks, the planner stabilizes inflation relative to the output gap on the margin more

Analogy with Calvo, 1983



State-dependent inflation-output tradeoff

- ▶ Inflation-output tradeoff varies with frequency
- ▶ After large shocks, the planner stabilizes inflation relative to the output gap on the margin more Analogy with Calvo, 1983
- ▶ Reduction in sacrifice ratio dominates decline in relative welfare weight of inflation



Robustness: alternative state-dependent models

- ▶ Menu cost is a random variable: (i) Calvo Plus (Nakamura and Steinsson, 2010),
(ii) Random Menu Cost (Dotsey et al., 1999; Gagliardone et al., 2025)

$$(i) \quad \tilde{\eta} = \begin{cases} \eta & \text{with prob } \alpha \\ 0 & \text{with prob } 1 - \alpha \end{cases} \quad (ii) \quad \tilde{\eta} = \begin{cases} U[-\eta, \eta] & \text{with prob } \alpha \\ 0 & \text{with prob } 1 - \alpha \end{cases}$$

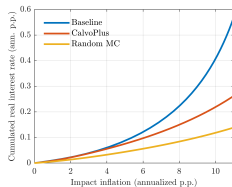
Robustness: alternative state-dependent models

- ▶ Menu cost is a random variable: (i) Calvo Plus (Nakamura and Steinsson, 2010), (ii) Random Menu Cost (Dotsey et al., 1999; Gagliardone et al., 2025)

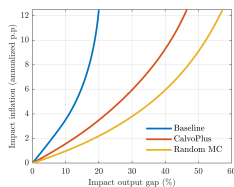
$$(i) \quad \tilde{\eta} = \begin{cases} \eta & \text{with prob } \alpha \\ 0 & \text{with prob } 1 - \alpha \end{cases} \quad (ii) \quad \tilde{\eta} = \begin{cases} U[-\eta, \eta] & \text{with prob } \alpha \\ 0 & \text{with prob } 1 - \alpha \end{cases}$$

- ▶ Strike-it relationship robust, weaker in alternatives, which underestimate frequency

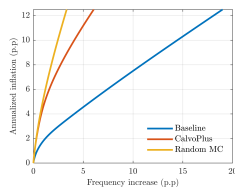
(a) Strike-it relationship



(b) Phillips relationship



(c) Frequency



Robustness: Sensitivity to parameter choice

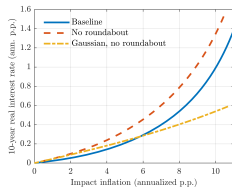
- ▶ Key components

- ▶ Roundabout economy (labor share, $\alpha = 0.2$): More realistic slope of the Phillips curve
- ▶ Fat tailed shocks (relative stddev, $\nu = 0.0625$): More realistic frequency response

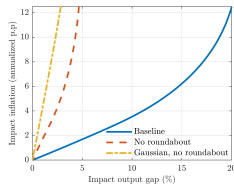
Robustness: Sensitivity to parameter choice

- ▶ Key components
 - ▶ Roundabout economy (labor share, $\alpha = 0.2$): More realistic slope of the Phillips curve
 - ▶ Fat tailed shocks (relative stddev, $\nu = 0.0625$): More realistic frequency response
- ▶ Strike-it relationship robust, weaker w/o fat tailed shocks, which underestimate frequency

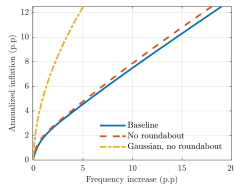
(a) Strike-it relationship



(b) Phillips relationship



(c) Frequency



Robustness: Calibration

	Baseline	CalvoPlus	Random MC	Description	Reference
<i>Households and final producer</i>					
β	$0.99^{1/3}$	$0.99^{1/3}$	$0.99^{1/3}$	Discount rate	Golosov and Lucas (2007)
χ	1	1	1	Utility weight on labor	Set so that $w = C$
ϵ	6	6	6	Elasticity of substitution	Gagliardone et al. (2025)
<i>Differentiated goods producers</i>					
α	0.2	0.2	0.2	Labor share	Nakamura and Steinsson (2010)
τ	0.1672	0.1685	0.1709	Labor subsidy in ss	$y = y^*$ in ss
θ	0	8.7%/2	8.7%/2	Share of 0 menu cost in ss	As in 2-product firms in Midrigan (2011)
η	0.064	0.468	2.40	Menu cost	Calibrated to match
σ	0.0306	0.0328	0.0365	Std dev. of quality shocks	moments of steady state
σ_1/σ_2	0.065	0.065	0.065	Ratio of low vs. high stdev	price change distribution
ϖ	0.934	0.926	0.9095	Share of low volat. shock	at 2% inflation
<i>Aggregate shocks</i>					
ρ_A	$0.95^{1/3}$	$0.95^{1/3}$	$0.95^{1/3}$	TFP shock persistence	Smets and Wouters (2007)
ρ_τ	$0.9^{1/3}$	$0.9^{1/3}$	$0.9^{1/3}$	Cost-push persistence	Smets and Wouters (2007)

Robustness: Calibration

	Baseline	$\alpha = 1$	$\alpha = 1, \nu = 1$	Description	Reference
<i>Households and final producer</i>					
β	$0.99^{1/3}$	$0.99^{1/3}$	$0.99^{1/3}$	Discount rate	Golosov and Lucas (2007)
χ	1	1	1	Utility weight on labor	Set so that $w = C$
ϵ	6	6	6	Elasticity of substitution	Gagliardone et al. (2025)
<i>Differentiated goods producers</i>					
α	0.2	1	1	Labor share	Nakamura and Steinsson (2010)
η	0.064	0.013	0.035	Menu cost	Calibrated to match
σ	0.0306	0.0306	0.0252	Std dev. of quality shocks	moments of steady state
σ_1/σ_2	0.065	0.065	1	Ratio of low vs. high stdev	price change distribution
ϖ	0.934	0.934	0.934	Share of low volat. shock	at 2% inflation
<i>Aggregate shocks</i>					
ρ_A	$0.95^{1/3}$	$0.95^{1/3}$	$0.95^{1/3}$	TFP shock persistence	Smets and Wouters (2007)
ρ_τ	$0.9^{1/3}$	$0.9^{1/3}$	$0.9^{1/3}$	Cost-push persistence	Smets and Wouters (2007)

Robustness: Calibration, cont.

	Steady state at 2% inflation				Phillips curve	10% inflation	
	Frequency	Size	Kurtosis	IQR	slope	Freq. incr.	Shock size
Data	8.7%	8.5%	4	10.8%	0.005-0.05	14pp	
Baseline	8.7%	8.5%	4	7.2%	0.08	14pp	-2.89%
CalvoPlus	8.7%	8.5%	4	10.8%	0.036	4.2pp	-3.32%
Random MC	8.7%	8.5%	4	10.5%	0.040		

Robustness: Calibration, cont.

	Steady state at 2% inflation			Phillips curve	10% inflation	
	Frequency	Size	Kurtosis	slope	Freq. incr.	Shock size
Data	8.7%	8.5%	4	0.005-0.05	14pp	
Baseline	8.7%	8.5%	4	0.08	14pp	-2.89%
$\alpha = 1$	8.7%	8.5%	4	0.42	13.72pp	-14.7%
$\alpha = 1, \nu = 1$	8.7%	8.5%	1.75	1.13	3.32pp	-6.15%

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